

Supplementary Table – Calibration constraints used for the 325–species DTOL chronogram (all 64 nodes)

Min/Max = calibration constraint bounds imposed on chronos. Calib. age = calibrated node age (chronos correlated). TT median / TT range = TimeTree pairwise median and range (API). Source: TimeTree (62) or literature (2, italic).

Clade	Node	Taxon A	Taxon B	Min (Ma)	Max (Ma)	Calib. age	TT median	TT adjusted	TT range	Source
Root	Eukaryota root	Eimeria tenella	Taurulus bubalis	1085.0	1671.0	1353.6	1530.0	1598.4	1085.0...1671.0	TimeTree
Opisthokonta	Opisthokonta	Membranipora membranacea	Agaricus bisporus	985.0	1392.0	985.0	1274.6	1274.6	985.0...1392.0	TimeTree
Metazoa	Falco	Falco peregrinus	Falco cherrug	1.7	2.5	2.5	2.1	2.1	1.7...2.8	TimeTree
Metazoa	Thunnus	Thunnus albacares	Thunnus maccoyii	3.0	4.4	3.1	3.9	3.7	3.4...4.3	TimeTree
Metazoa	Patella	Patella vulgata	Patella pellucida	7.2	10.8	10.8	9.0	...	0.0...0.0	TimeTree
Metazoa	Limnephilidae (caddisfly)	Glyphotaelius pellucidus	Limnephilus lunatus	10.0	18.0	10.0	...	...	...	Lit: 2026 Limnephilidae phylogenomics study
Metazoa	Taeniopygia/Corvus	Taeniopygia guttata	Corvus hawaiiensis	12.8	19.2	19.2	38.0	16.0	0.0...38.0	TimeTree
Metazoa	Tridacna	Tridacna crocea	Tridacna gigas	17.0	147.1	47.2	82.1	51.6	17.1...147.2	TimeTree
Metazoa	Polyneoptera	Leuctra nigra	Ectobius pallidus	25.0	370.6	271.9	270.4	308.1	25.0...370.6	TimeTree
Metazoa	Lagopus/Gallus	Lagopus muta	Gallus gallus	28.6	43.0	39.0	32.8	35.8	27.9...42.4	TimeTree
Metazoa	Phorcus/Steromphala	Phorcus lineatus	Steromphala cineraria	37.4	56.0	37.4	76.0	46.7	0.0...0.0	TimeTree
Metazoa	Porphyrio/Lagopus	Porphyrio hochstetteri	Lagopus muta	72.6	109.0	72.6	90.8	...	86.5...101.8	TimeTree
Metazoa	Leistus/Pterostichus	Leistus spinibarbis	Pterostichus madidus	81.4	122.0	122.0	45.7	101.7	0.0...0.0	TimeTree
Metazoa	Apidae	Bombus hypnorum	Nomada fabriciana	98.4	112.0	98.4	108.0	108.0	98.5...112.0	TimeTree
Metazoa	Otophysi	Danio rerio	Pangasianodon hypophthalmus	132.0	170.0	132.0	142.1	...	132.0...170.0	TimeTree
Metazoa	Avemetatarsalia/Ornithodira node	Micropterix aruncella	Atherix ibis	151.9	344.7	341.2	286.1	299.4	151.9...344.7	TimeTree
Metazoa	Paraneoptera	Icerya purchasi	Mesopsocus fuscifrons	172.6	416.6	381.2	353.1	...	172.6...416.6	TimeTree
Metazoa	Coleoptera	Chrysolina oricalcia	Ophonus ardosiacus	189.6	307.2	291.6	279.6	282.4	189.6...307.2	TimeTree
Metazoa	Amphiesmenoptera	Pieris napi	Limnephilus lunatus	200.0	290.0	261.5	246.4	246.4	172.3...302.4	TimeTree
Metazoa	Aculeata/Tipuloidea node	Abia candens	Tiphia femorata	223.7	304.0	223.7	239.5	264.8	223.7...304.0	TimeTree
Metazoa	Sauria	Lagopus muta	Vipera latastei	274.9	286.8	274.9	279.8	...	274.9...286.8	TimeTree
Metazoa	Endopterygota	Anoplius nigerrimus	Leistus spinibarbis	310.7	389.7	377.4	344.0	...	310.7...389.7	TimeTree
Metazoa	Amniota	Diceros bicornis	Lagopus muta	316.0	322.4	316.0	318.9	...	316.0...322.4	TimeTree
Metazoa	Palaeoptera	Siphonurus alternatus	Ischnura elegans	325.8	430.6	377.3	365.5	...	325.8...430.6	TimeTree
Metazoa	Actinopteri	Acipenser ruthenus	Thunnus maccoyii	333.4	372.0	333.4	349.9	...	333.4...372.0	TimeTree
Metazoa	Neoptera	Leuctra nigra	Nephrotoma flavescens	346.5	391.8	391.8	383.1	...	346.5...391.8	TimeTree
Metazoa	Pterygota	Nephrotoma flavescens	Ischnura elegans	377.4	441.6	420.2	401.0	...	377.4...441.6	TimeTree
Metazoa	Gastropoda	Steromphala cineraria	Patella pellucida	379.7	500.0	379.7	421.6	476.3	379.7...500.0	TimeTree
Metazoa	Euteleostomi	Acipenser ruthenus	Falco cherrug	423.3	440.0	423.3	429.0	...	423.2...440.0	TimeTree
Metazoa	Hexapoda	Nephrotoma flavescens	Orchesella flavescens	425.4	478.1	478.1	459.0	459.0	425.4...478.1	TimeTree
Metazoa	Arthropoda	Nephrotoma flavescens	Meta bourneti	536.6	600.0	545.6	547.6	563.3	536.6...600.0	TimeTree
Metazoa	Protostomia	Membranipora membranacea	Nephrotoma flavescens	543.0	670.1	606.7	610.0	691.5	543.0...670.1	TimeTree
Metazoa	Bilateria	Marthasterias glacialis	Nephrotoma flavescens	627.0	830.0	629.7	686.2	707.6	627.0...830.0	TimeTree
Metazoa	Metazoa	Nephrotoma flavescens	Sycon ciliatum	701.8	1020.0	736.1	758.1	...	701.9...1020.0	TimeTree
Viridiplantae	Arabidopsis	Arabidopsis thaliana	Cardamine hirsuta	5.9	29.2	29.2	17.1	18.4	5.9...29.2	TimeTree
Viridiplantae	Luzula	Luzula pallescens	Luzula sylvatica	6.0	10.0	10.0	...	...	...	Lit: Luzula crown ~10 Ma (literature)
Viridiplantae	Geum	Geum rivale	Geum urbanum	10.3	15.5	15.5	12.9	...	0.0...0.0	TimeTree
Viridiplantae	Potamogetonaceae/Droseraceae	Rosa canina	Potentilla anserina	21.1	57.8	32.1	27.1	39.5	21.1...57.8	TimeTree
Viridiplantae	Hologalegina	Trifolium fragiferum	Astragalus glycyphyllos	27.8	44.3	44.3	35.2	34.2	27.8...44.2	TimeTree
Viridiplantae	Cyperaceae clade	Schoenoplectus triqueter	Carex depauperata	29.9	59.6	48.9	50.7	...	29.9...59.6	TimeTree
Viridiplantae	Asteraceae node	Tanacetum vulgare	Arctium minus	36.6	45.1	45.1	39.1	42.4	36.6...45.1	TimeTree
Viridiplantae	Poaceae/Asteraceae node	Phragmites australis	Hierochloe odorata	41.4	51.9	51.9	49.1	59.1	41.4...51.9	TimeTree
Viridiplantae	Juncaceae	Luzula sylvatica	Juncus squarrosus	44.3	69.6	65.0	54.6	65.7	44.3...69.6	TimeTree
Viridiplantae	Rhynchoszegium/Thuidium (moss)	Rhynchoszegium confertum	Thuidium tamariscinum	63.2	94.8	63.2	71.7	79.0	48.8...94.6	TimeTree
Viridiplantae	Rosales	Geum urbanum	Urtica urens	76.0	95.5	95.5	87.6	...	76.0...95.5	TimeTree
Viridiplantae	Caryophyllales	Polygonum aviculare	Arenaria serpyllifolia	81.3	97.0	97.0	86.1	106.3	81.3...97.0	TimeTree
Viridiplantae	Poales	Sparganium emersum	Juncus squarrosus	94.1	117.0	117.0	106.0	110.1	94.1...117.0	TimeTree
Viridiplantae	Pentapetalae	Geum urbanum	Polygonum aviculare	111.4	123.9	123.9	118.0	125.1	111.4...123.9	TimeTree
Viridiplantae	Alismatales/Poales node	Potamogeton natans	Juncus squarrosus	125.3	137.9	137.9	132.4	138.8	125.3...137.9	TimeTree
Viridiplantae	Fabales/Caryophyllales node	Trifolium fragiferum	Berberis vulgaris	126.0	132.4	132.4	129.3	142.2	126.0...132.4	TimeTree
Viridiplantae	Caryophyllales/Poales node	Polygonum aviculare	Juncus squarrosus	142.1	163.5	163.5	159.6	159.6	142.1...163.5	TimeTree
Viridiplantae	Moss node	Polytrichastrum sexangulare	Rhynchoszegium confertum	181.3	354.8	181.3	213.9	230.4	181.3...354.9	TimeTree
Viridiplantae	Liverwort node	Herbertus hutchinsiae	Lunularia cruciata	304.8	437.9	304.8	374.2	376.4	304.8...437.9	TimeTree
Viridiplantae	Embryophyta	Herbertus hutchinsiae	Rosa canina	460.0	497.3	460.0	479.8	483.7	460.0...497.3	TimeTree
Viridiplantae	Chlorophyceae	Dunaliella primolecta	Marvania coccoides	662.0	1000.0	662.0	827.5	...	662.0...1000.0	TimeTree
Viridiplantae	Viridiplantae	Rosa canina	Pycnococcus provasolii	729.0	1105.7	871.8	936.3	991.8	729.0...1105.7	TimeTree
Fungi	Panellus Mycena	Mycena galericulata	Panellus stipticus	2.1	13.7	13.7	7.9	4.0	2.1...13.7	TimeTree
Fungi	Agaricus Chlorophyllum	Chlorophyllum brunneum	Agaricus bisporus	46.4	73.3	73.3	59.8	59.0	46.4...73.3	TimeTree
Fungi	Clavariaceae	Ramariopsis corniculata	Clavaria fumosa	56.5	69.3	69.3	62.9	62.9	56.5...69.3	TimeTree
Fungi	Russula Hericium	Hericium erinaceus	Russula vesca	69.0	143.1	143.1	114.2	111.8	69.0...143.1	TimeTree
Fungi	Platismatia Cladonia	Cladonia squamosa	Platismatia glauca	93.9	146.4	104.5	139.0	124.7	93.9...146.4	TimeTree
Fungi	Sordariomyceta	Leotia lubrica	Cordyceps fumosorosea	239.7	320.0	272.9	293.5	320.4	239.7...320.0	TimeTree
Fungi	Saccharomyceta	Metschnikowia zobellii	Platismatia glauca	437.8	670.2	609.9	533.6	523.5	437.8...670.2	TimeTree
Fungi	Dikarya	Agaricus bisporus	Platismatia glauca	583.2	749.0	728.4	642.0	...	583.2...749.0	TimeTree